Annexes : Tutoriel de tentatives d’intrusion via Kali et Ettercap

SecureTech Innovation

AISSAM ILYESS

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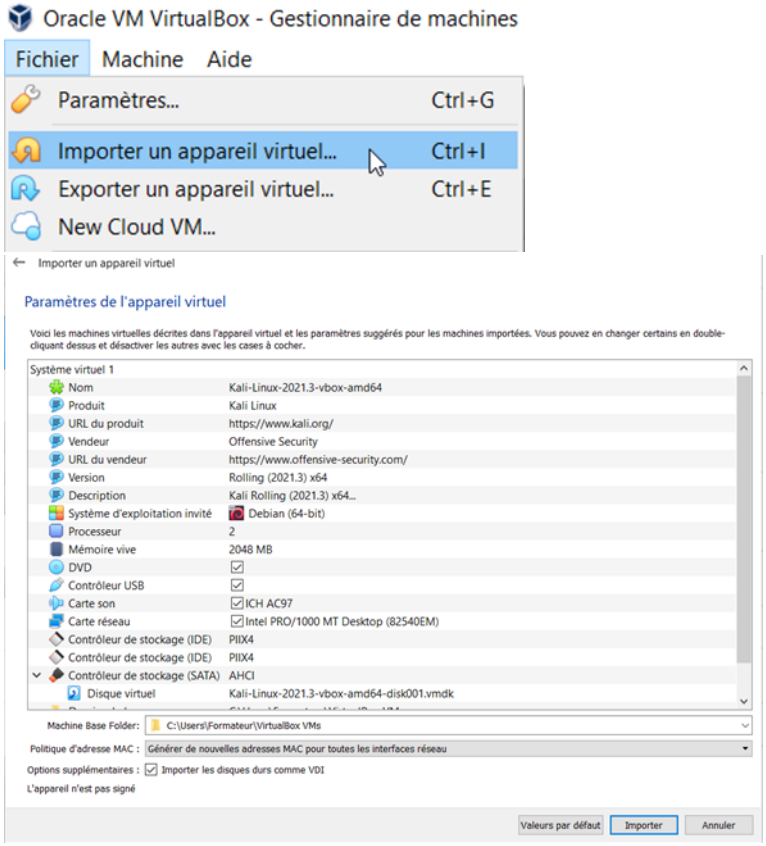
Configuration Pfsense

Empoisonnement cache ARP

Linux Kali (hacker)

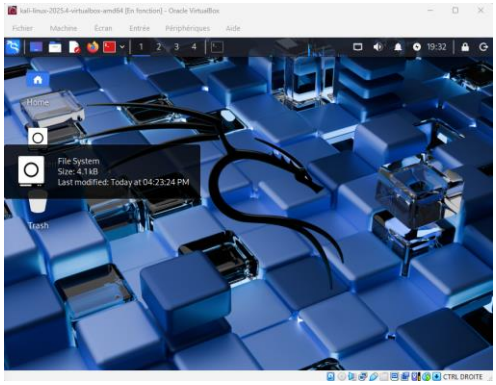
Télécharger Kali : https://www.kali.org/get-kali/

• Importer un OVA



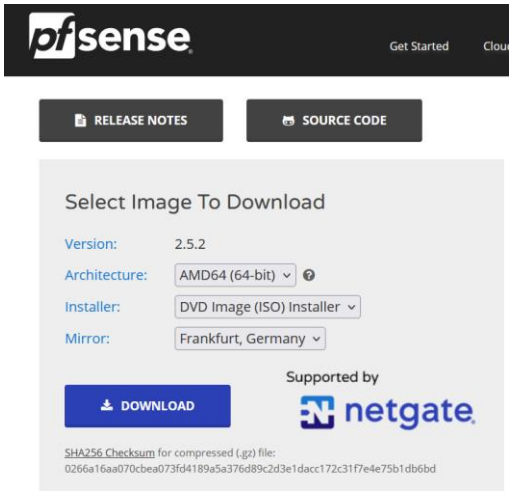
Config Kali :

- Clavier azerty: **setxkbmap fr**
- Mise à jour : **apt-get update && apt upgrade**
- Nouveau mot de passe root : **passwd ...**



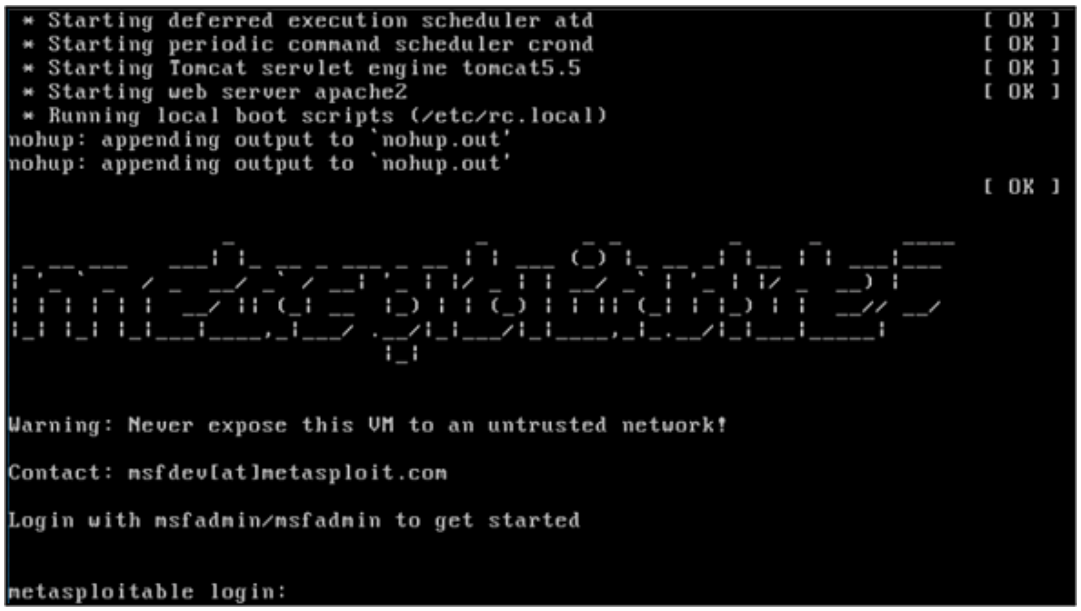
Pfsense (firewall)

Télécharger Pfsense : https://www.pfsense.org/download/



Metasploit (machine vulnérable)

Télécharger metasploitable : https://sourceforge.net/projects/metasploitable/



Client légitime (Linux ou Windows)

Client victime debian : Le client légitime pourra être soit une VM Ubuntu soit une VM Windows 7

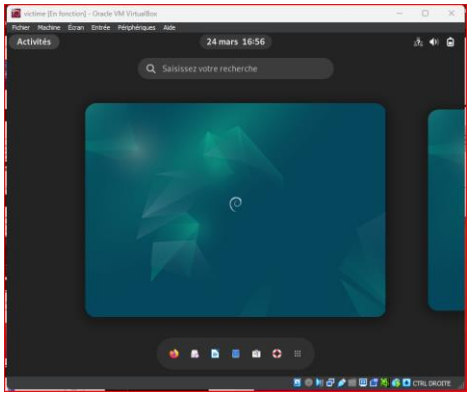
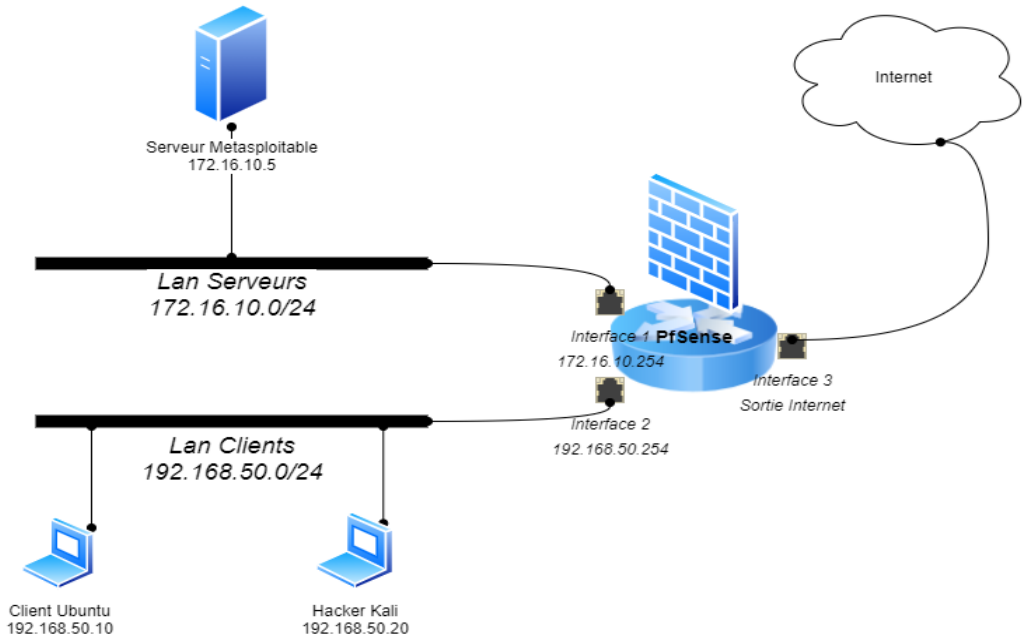


Schéma et Plan d’adressage

Schéma infrastructure :



Plan d'adressage IP :

Machines	Descriptions	IP	Passerelle
Hacker	Machine virtuelle Kali Linux	192.168.50.20/24	192.168.50.254
Firewall	Firewall pfsense	interface 1 : 172.16.10.254 interface 2 : 192.168.50.254	Interface 3 : IP Dynamique sortie internet via le réseau de l'UTEC
Serveur Mestasploitable	Machine virtuelle vulnérable	172.16.10.5/24	172.16.10.254
Client légitime	Machine linux ou windows avec un navigateur	192.168.50.10/24	192.168.50.254

Configuration réseau :

Pour toutes les VMs :

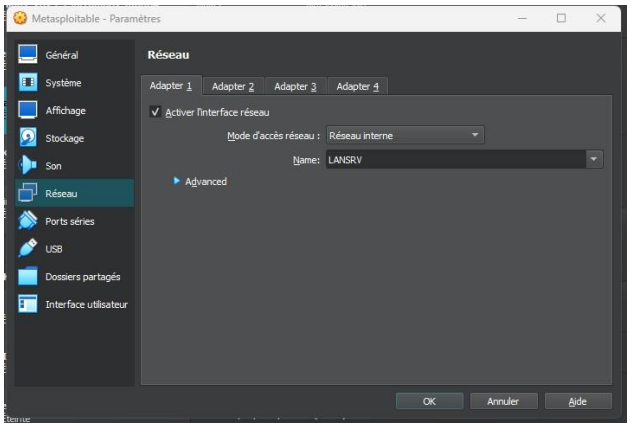
- Activer l'interface réseau

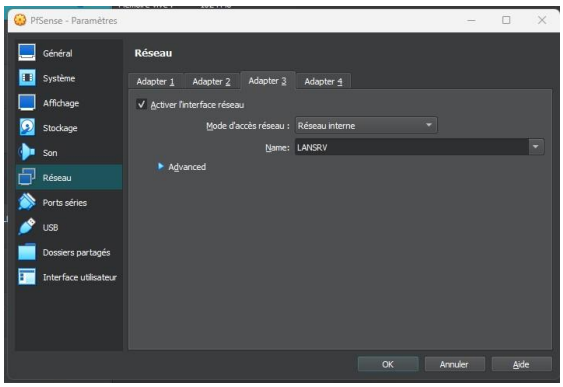
•Mode d’accès réseau : “**Réseau interne**”

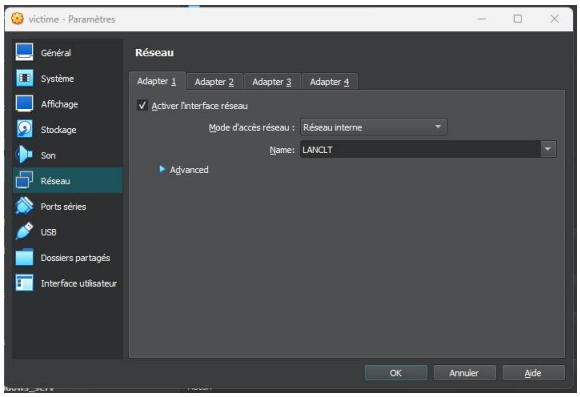
•Mode de promiscuité : “**Allow All**”

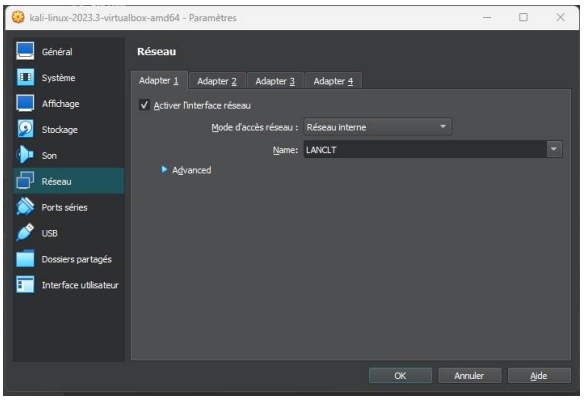
•Adresse MAC : **<u>unique</u> sinon générer une nouvelle**

•Cable branché

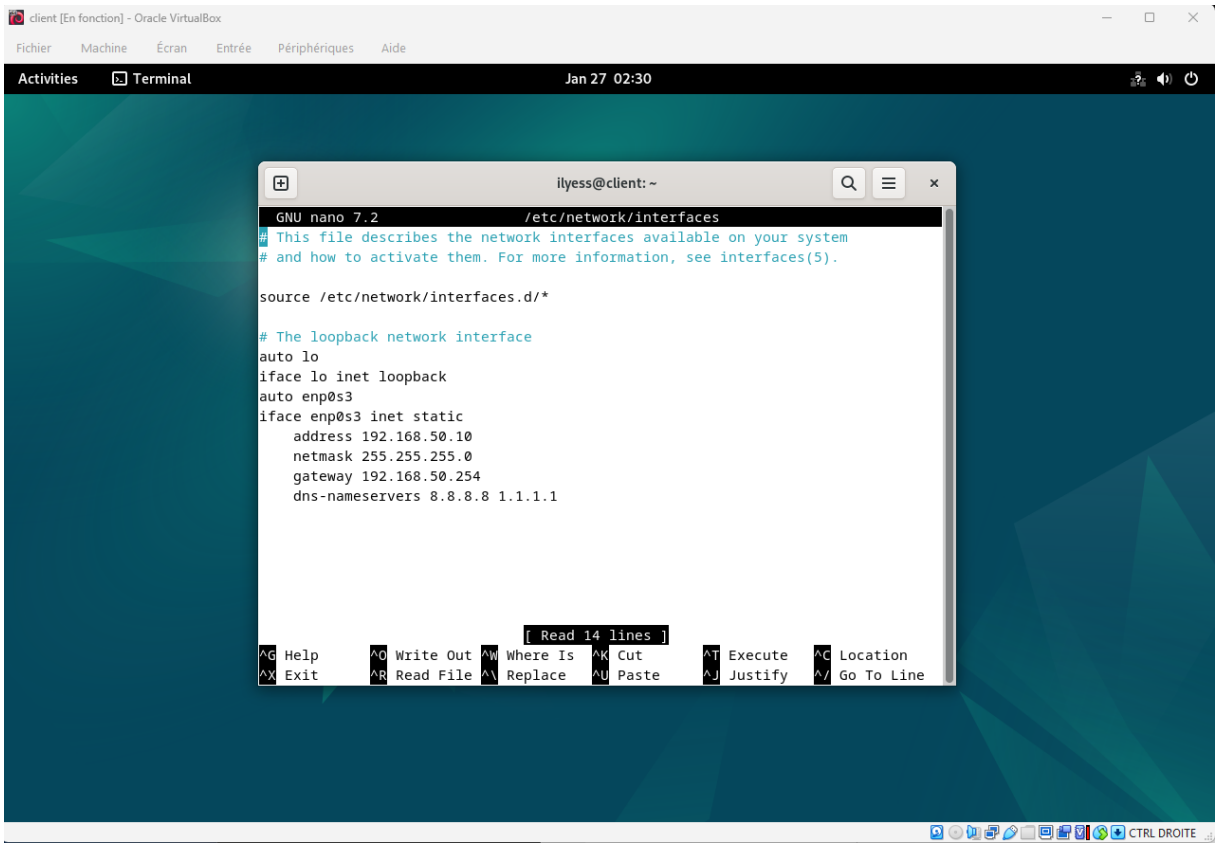




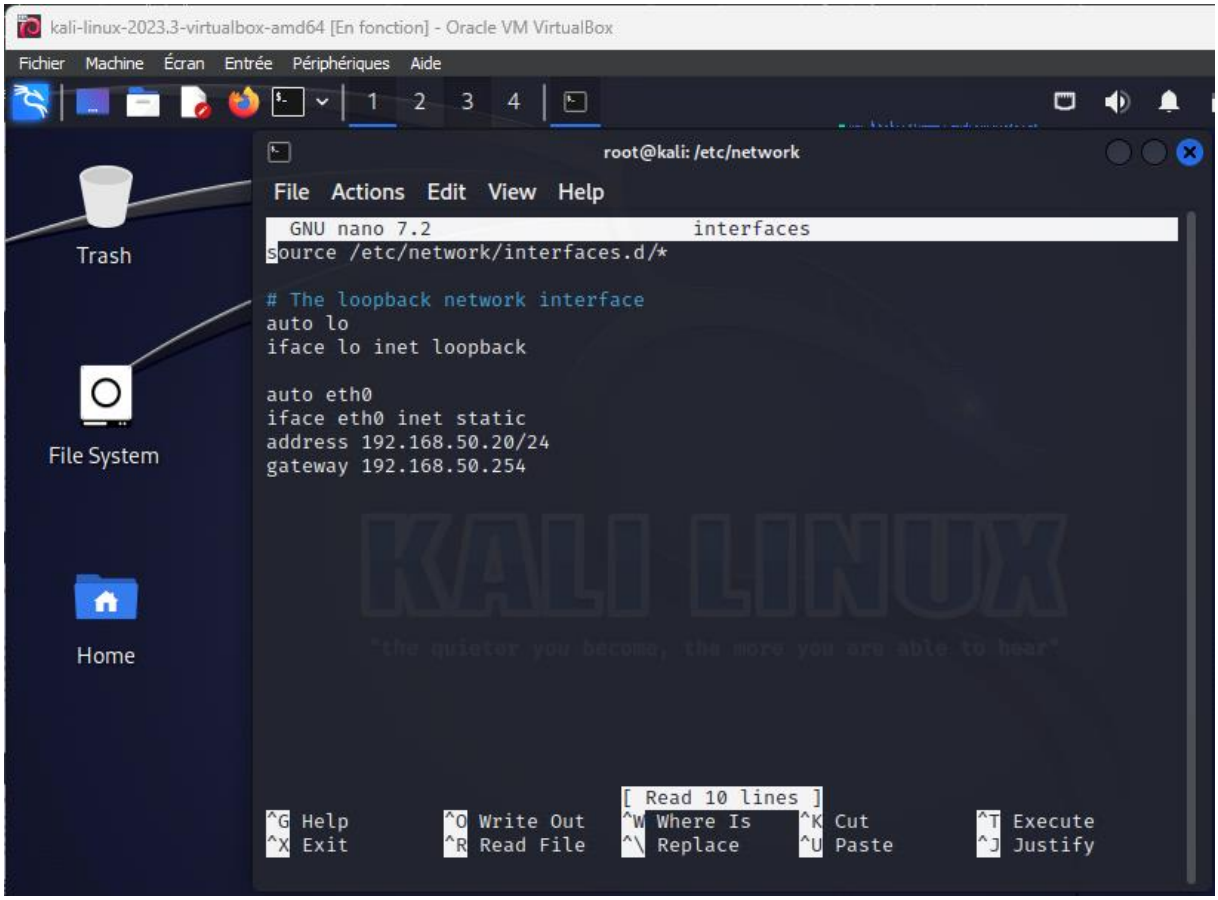




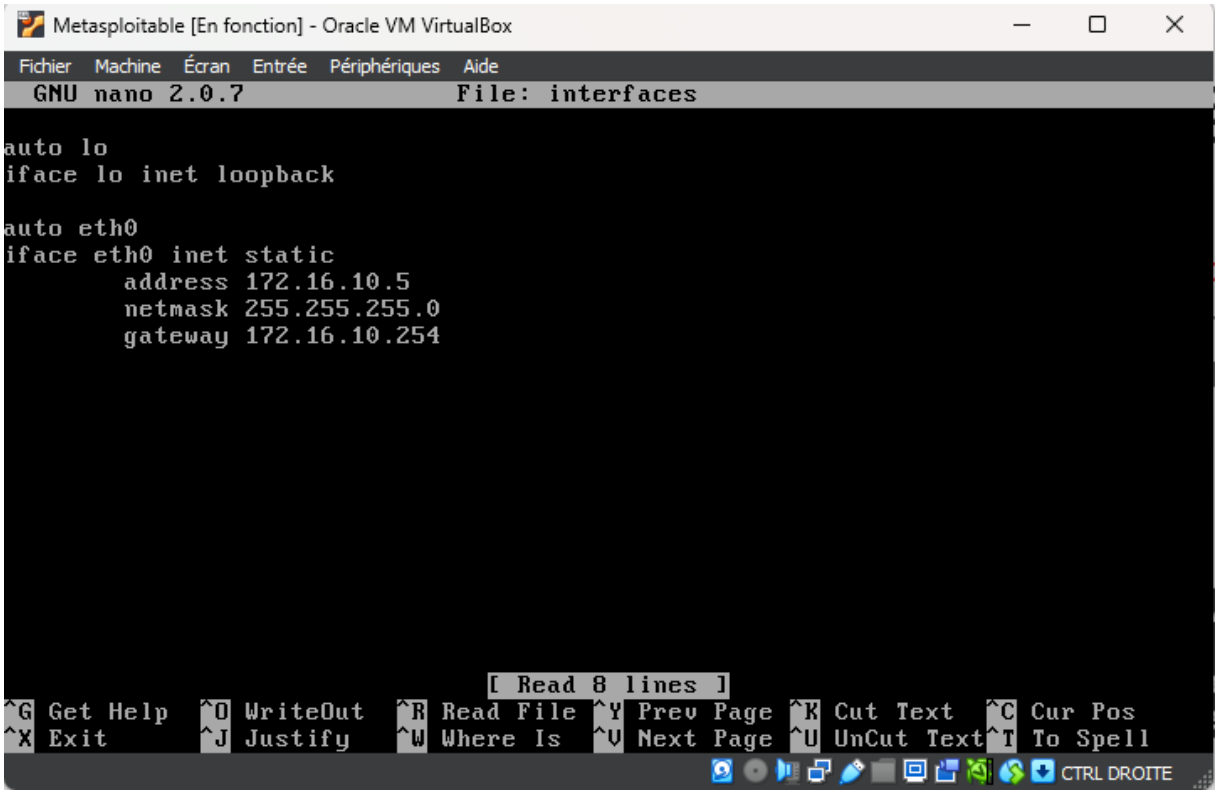
Config réseau Victime :


client [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
Activities Terminal Jan 27 02:30
ilyess@client: ~
GNU nano 7.2 /etc/network/interfaces
This file describes the network interfaces available on your system
and how to activate them. For more information, see interfaces(5).
source /etc/network/interfaces.d/*
The loopback network interface
auto lo
iface lo inet loopback
auto enp0s3
iface enp0s3 inet static
address 192.168.50.10
netmask 255.255.255.0
gateway 192.168.50.254
dns-nameservers 8.8.8.8 1.1.1.1
[Read 14 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line
CTRL DROITE

Config Kali :


kali-linux-2023.3-virtualbox-amd64 [En fonction] - Oracle VM VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
1 2 3 4
root@kali: /etc/network
File Actions Edit View Help
Trash
File System
Home
GNU nano 7.2 interfaces
source /etc/network/interfaces.d/*
The loopback network interface
auto lo
iface lo inet loopback
auto eth0
iface eth0 inet static
address 192.168.50.20/24
gateway 192.168.50.254
KALI LINUX
"the quieter you become, the more you are able to hear"
[Read 10 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

Config metasploitable :



Nous allons maintenant configurer pfsense

Activer **Adaptateur 1** de VirtualBox :

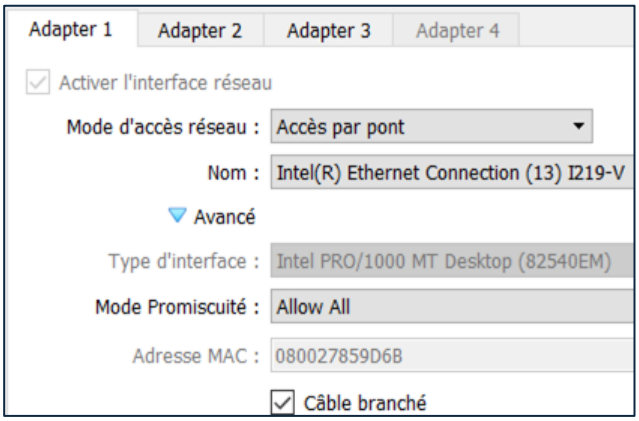
✓ Activer l'interface réseau

Mode d'accès réseau : "**Accès par pont**"

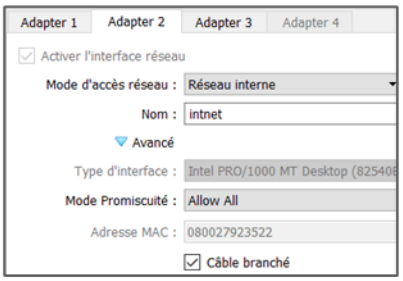
Mode de promiscuité : "**Allow All**"

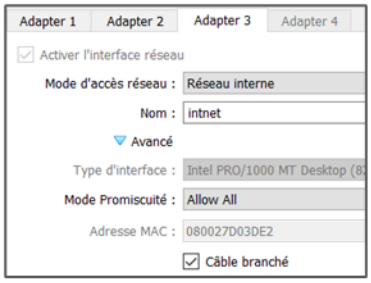
Adresse MAC : **<u>unique</u> sinon générer une nouvelle**

✓ Cable branché



- ❑ Activer **Adaptateur 2 et 3** avec la configuration





- ❑ **✓ Activer l'interface réseau**
- ❑ Mode d'accès réseau : "**Réseau interne**"
- ❑ Nom : "**intnet**"
- ❑ Mode de promiscuité : "**Allow All**"
- ❑ Adresse MAC : **<u>unique</u> sinon générer une nouvelle**
- ❑ **✓ Cable branché**

Configuration Pfsense

1) Assign Interfaces

- ❑ Do you want to set up VLANs now [y|n]? **n**
- ❑ Enter the **WAN** interface name or ‘a’ for auto-detection: **em0**
- ❑ Enter the **LAN** interface name or ‘a’ for auto-detection: **em1**
- ❑ Enter the **Optional 1** interface name or ‘a’ for auto-detection: **em2**
- ❑ Enter the Optional 2 interface name or ‘a’ for auto-detection (or nothing if finished) : **faire entrée** ↲
- ❑ Do you want to proceed [y|n]? **y**

```
Do you want to set up VLANs now [y|n]? n

*NOTE*  pfSense requires *AT LEAST* 1 assigned interface(s) to function.
        If you do not have *AT LEAST* 1 interfaces you CANNOT continue.

        If you do not have at least 1 *REAL* network interface card(s)
        or one interface with multiple VLANs then pfSense
        *WILL NOT* function correctly.

If you do not know the names of your interfaces, you may choose to use
auto-detection. In that case, disconnect all interfaces now before
hitting 'a' to initiate auto detection.

Enter the WAN interface name or 'a' for auto-detection: em0

Enter the LAN interface name or 'a' for auto-detection
NOTE: this enables full Firewalling/NAT mode.
(or nothing if finished): em1

Enter the Optional 1 interface name or 'a' for auto-detection
(or nothing if finished): em2

Enter the Optional 2 interface name or 'a' for auto-detection
(or nothing if finished):
```

```
The interfaces will be assigned as follows:

WAN  -> em0
LAN  -> em1
OPT1 -> em2

Do you want to proceed [y|n]?y
```

2) Set Interface(s) IP address

- ❑ Enter the number of the interface you wish to configure : **2**
- ❑ Enter the new LAN IPV4 address : **192.168.50.254**
- ❑ Enter the new LAN IPV4 subnet bit count : **24**
- ❑ Enter the new LAN IPV4 gateway address : **faire entrée** ↲
- ❑ Enter the new LAN IPV6 address : **faire entrée** ↲
- ❑ Do you want to enable the DHCP server on LAN [y|n] : **n**

Do you want to revert to HTTP as the webConfigurator protocol [y|n] : **y**

```
Do you want to enable the DHCP server on LAN? [y|n]  n
Disabling DHCPD...Done!
Disabling DHCPD...Done!

Do you want to revert to HTTP as the webConfigurator protocol? (y/n) y

Please wait while the changes are saved to LAN... Reloading filter...
 DHCPD... restarting webConfigurator...

The IPv4 LAN address has been set to 192.168.50.254/24
You can now access the webConfigurator by opening the following URL in your web
browser:
                http://192.168.50.254/

Press <ENTER> to continue.
```

Lancer le navigateur, entrez l’adresse IP du firewall et se connecter :

Renommer les interfaces

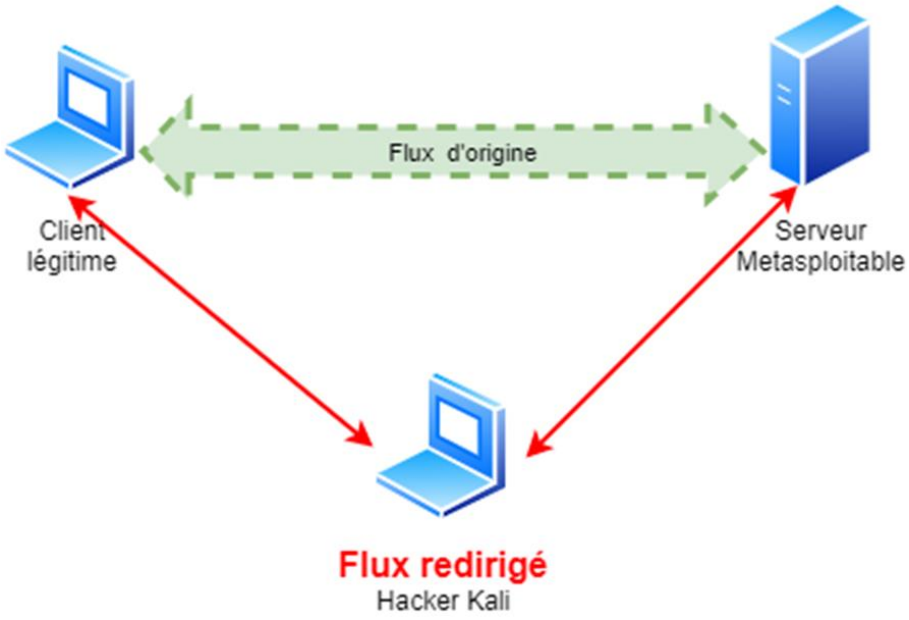
LAN -> LAN_CLIENTS

OPT1 -> LAN_SERVERS

WAN -> WAN

Empoisonnement cache ARP

ARP POISONING :



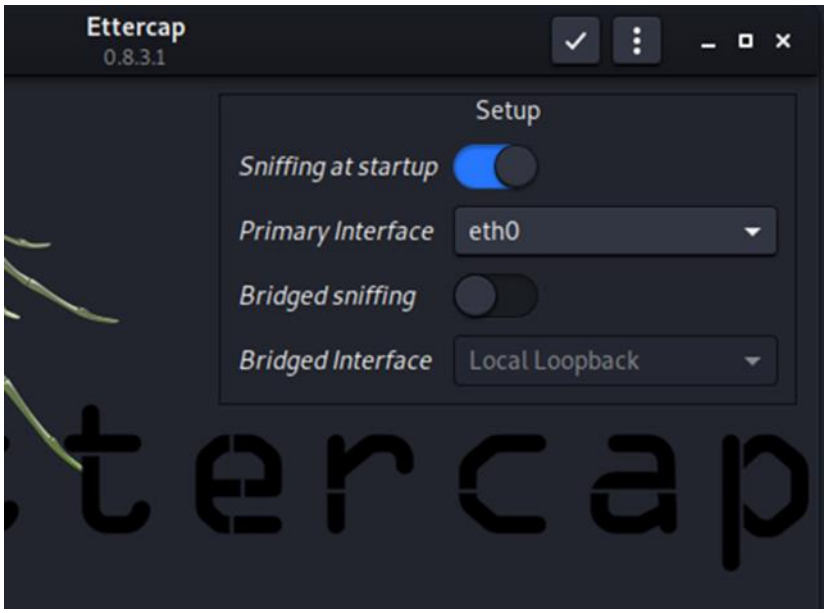
Scanner le reseau :

- Notre IP -> ifconfig
- Notre Gateway -> ip route show
- Les IPs des autres machines :
    - netdiscover
    - nmap 192.168.50.0/24

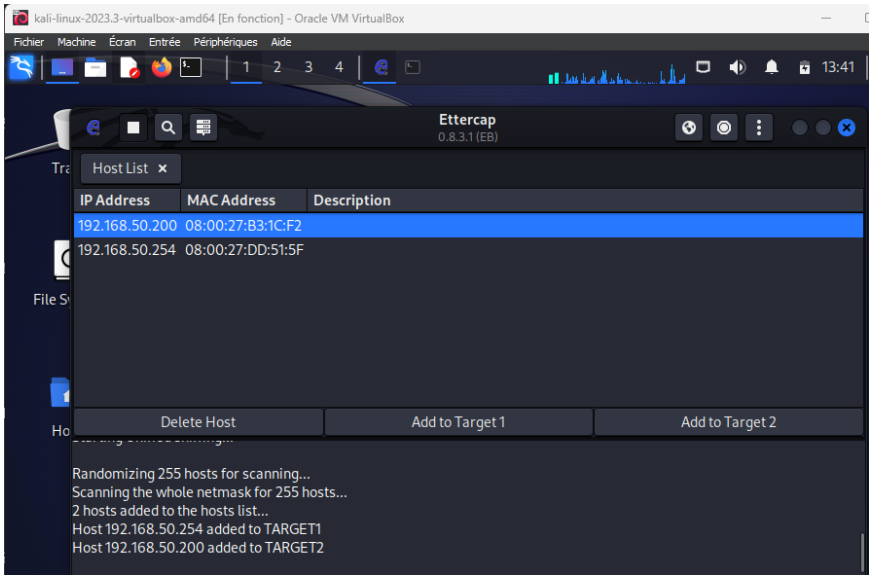
```
┌──(root㉿kali)-[/etc/network]
└─# nmap 192.168.50.0/24
Starting Nmap 7.94 ( https://nmap.org ) at 2024-03-24 13:36 EDT
Nmap scan report for 192.168.50.200
Host is up (0.00094s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
10000/tcp open  snet-sensor-mgmt
MAC Address: 08:00:27:B3:1C:F2 (Oracle VirtualBox virtual NIC)

Nmap scan report for 192.168.50.254
Host is up (0.00090s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT   STATE SERVICE
53/tcp open  domain
80/tcp open  http
MAC Address: 08:00:27:DD:51:5F (Oracle VirtualBox virtual NIC)

Nmap scan report for 192.168.50.20
Host is up (0.0000010s latency).
All 1000 scanned ports on 192.168.50.20 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 256 IP addresses (3 hosts up) scanned in 36.95 seconds
```

Empoisonnement ettercap :

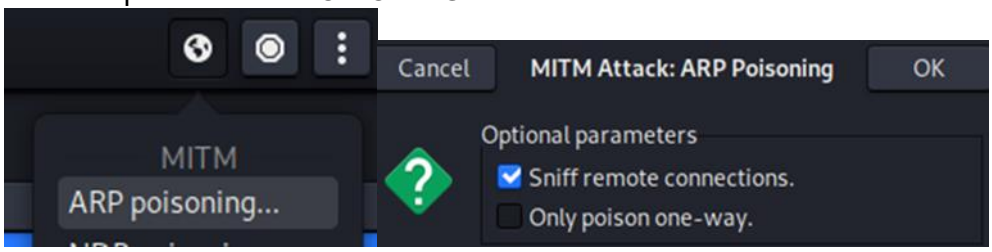
1. Interface eth0
2. Accept



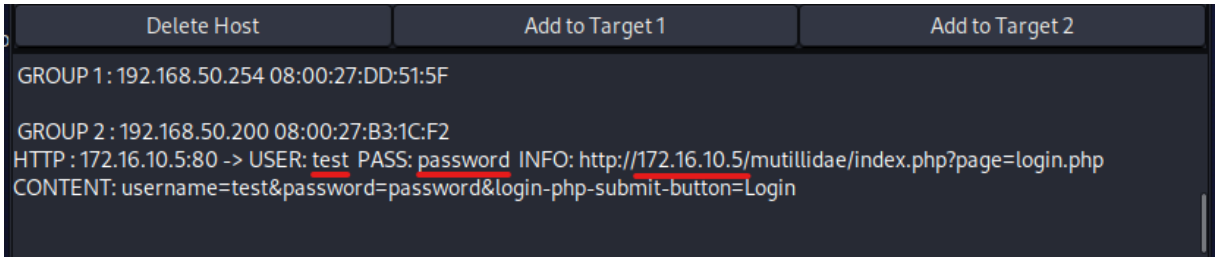
Apres une recherche du réseau 2 adresse IP remonte, la gateway et la victime.



On ajoute la gateway a la target 1 et la victime a la target 2, on va ensuite dans le menu MITM cliquer sur ARP POISONING et activer « Sniff remote connections. »



Depuis la victime on se connecte sur le portail captif du serveur metasploitable :

On utilisera ici le user test et le mot de passe password.

On se connecte, et instanement, dans le Kali, nous avons la remonté depuis ettercap :